

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 4

26 JUNE 2025

Overview:

We learned about **GitHub**, which is a platform used to store and manage code online. It helps developers collaborate, track changes, and work together on projects. We also understood the difference between **Git** (a version control tool) and **GitHub** (a cloud-based hosting service). Along with this, we practiced some basic **Git commands** used to upload, update, and manage code from the local computer to GitHub.

Learning Objectives:

- Understand what Git and GitHub are.
- Learn how version control helps in managing code.
- Use basic Git commands to push code to GitHub.
- Create repositories and manage files online.

What is GitHub?

GitHub is a **web-based platform** used by developers to **store, manage, and share code online**. It works with **Git**, which is a version control system that tracks changes in files and helps manage code efficiently. GitHub makes it easier for individuals and teams to **collaborate on projects**, review code, and keep track of different versions. It also allows users to create **repositories (repos)** where they can push, pull, and update their code. Many open-source and professional projects are hosted on GitHub because it helps keep the work organized, backed up, and accessible from anywhere.

Basic Git/GitHub Commands:

- **git config:** It is used to set the basic configurations on Git like your name and email.

```
git config --global user.name " " : It is used to sets configuration values for your
username on git.
```

- **git config --global user.name " " :** It is used to sets configuration values for your username on git.

```
git config --global user.name "Jeeya Thapar" //Set global username
```

- **git config --global user.email " " :** It is used to set configuration values for your user email on git.

```
git config --global user.email jeeyathapar@gmail.com // Set global email
```

- **mkdir:** It is used to Create a directory if not created initially.

```
mkdir store // Create a new directory
```

- **cd:** It is used to go inside the directory and work on its contents.

```
cd store // Enter directory
```

- **git init :** It is used to create a local git repository for us in our store folder. This will help to manage the git commands for that repository.

```
git init // Initialize Git repo
```

- **git status:** It is used to see what's changed since the last commit. It shows all the files that have been added and modified and are ready to be committed and files that are untracked.

```
git status //Check repo status
```

- **git add Readme.txt:** To add a file Readme.txt to the staging area to track its changes.

```
git add Readme.txt // Add Readme.txt file to the staging area
```

- **git add:** It is used to add a specific list of files to the staging area.

```
git add // Add specific files to the staging area (must specify filenames)
```

- **git add *.txt:** It is used to add all text files of the current directory to the staging area.

```
git add *.txt // Stage all .txt files in the current directory
```

- **git remote add <address>** : It is used to add new remotes to our local repository for a particular git address.

```
git remote add <address>
```

- **git remove rm**: It is used to remove a remote from our local repository.

```
git remove rm
```

- **git clone <https://github.com/mdaan124/MyAlgorithms>** : It is used to clone or make a local copy of the global repository in your system (git clone command downloads the repository and creates a remote named origin which can be checked by the command – git remote -v).

```
git clone https://github.com/mdaan124/MyAlgorithms
```

- **git branch**: It is used to see all the branches present and current branches that we are working on.

```
git branch
```

- **ls**: It is used to see directories and files in the current directory.

```
Ls
```

- **git merge Testing** : It is used to merge the Testing branch with the master branch.

```
git merge Testing
```

- **git fetch**: It is used to fetch down any changes from the global repository to the current repository.

```
git fetch
```

Conclusion:

Working with GitHub felt very useful and practical. Learning the commands helped me understand how code is saved, updated, and shared in real-world projects. It's a great tool for keeping backup of code and working in teams. At first, the commands seemed a bit tricky, but with practice, they started making sense